Equal-combination response as a function of p

Model (paper, restricted to the two directions as the only features, weights w_1, w_2, in-plane angle φ, and $g \equiv \sin^p\alpha\,\|a\|^p$):

$$R(\varphi) = F\big(\,S(\varphi)\,\big), \qquad S(\varphi) = g\,\big[\,w_1\cos^p\varphi + w_2\sin^p\varphi\,\big].$$

The three measured conditions (at fixed perturbation angle α):

$$\underbrace{S_1 = g\,w_1}_{\varphi=0}, \qquad \underbrace{S_2 = g\,w_2}_{\varphi=\pi/2} \qquad \underbrace{S_{\text{mix}} = g\,(w_1+w_2)\,2^{-p/2}}_{\varphi=\pi/4,\ \cos\varphi=\sin\varphi=2^{-1/2}}.$$

Hence, independent of w_1, w_2, g:

$$\boxed{\frac{S_{\text{mix}}}{S_1+S_2} = 2^{-p/2}} \qquad \text{and, for } w_1=w_2: \quad \frac{S_{\text{mix}}}{S_{\text{axis}}} = 2^{\,1-p/2}.$$

So $2^{-p/2} \in (0, \frac{1}{2}]$ for $p \geq 2$, and $\to 0$ only as $p \to \infty$. Identifying $R \approx S$ (F locally linear), the proxy $R_{\text{mix}}/(R_1+R_2)$ *equals* $2^{-p/2}$: minimizing it $\Leftrightarrow$ pushing $p \to \infty$.

Why $R_{\text{mix}} \to 0$ is not the $p \to \infty$ box

The relation above fixes only the *single* diagonal point at the axis magnitude. The full $p \to \infty$ contour (the box) still *breaks* along the diagonal — at the corner. In normalized coordinates $(x, y) = (\sin\alpha\cos\varphi/\sin\theta_1,\ \sin\alpha\sin\varphi/\sin\theta_2)$ the superellipse is $|x|^p + |y|^p = 1$; the diagonal $(x = y)$ crosses it at

$$x = y = 2^{-1/p}, \qquad \text{so as } p\to\infty: \quad x = y \to 1,$$

i.e. the corner (θ_1, θ_2), at geodesic radius $\sqrt{2}\,\theta$ (equal intercepts). So even for $p = \infty$ the diagonal breaks at a *finite* angle $\sqrt{2}\,\theta$; for any finite p the diagonal-point ratio $2^{-p/2}$ is strictly positive.

The optimizer instead drives $R_{\text{mix}} \approx 0$ *sustained* far past $\sqrt{2}\,\theta$ (data along the diagonal out to $\gtrsim 60^\circ$ with $\theta \approx 22^\circ$, corner $\approx 31^\circ$). No $p \in (0, \infty]$ produces that: it is outside the superellipse family, not its $p \to \infty$ limit. The single-angle ratio cannot distinguish the box from beyond-the-box; only the full contour can.